

6221-21084-ITAI-1371

Group Members:

Edwin Marquez

Kendrick Robinson

Ali Shan

Fernando Reyes

Professor: Sitaram Ayyagari

Date: 10/06/2025

Houston Community College

Houston, Texas

Academic Reflective Journal

Edwin Marquez

Completing the Module 03 Lab helped me understand the full machine learning (ML) workflow and the differences between supervised, unsupervised, and reinforcement learning. At the beginning, I felt that the process was a little complex because it involved many stages such as data collection, preprocessing, model training, and evaluation. However, as I went through each of the 25 cells step by step, everything started to make more sense. I realized that ML is not only about building a model but also about understanding how every step contributes to the final performance.

One of the most valuable lessons for me was learning how to distinguish between supervised and unsupervised learning. Before this lab, I only had a basic idea of the difference, but now I understand it much better. Supervised learning uses labeled data — like having a teacher that provides the correct answers — while unsupervised learning deals with unlabeled data and looks for patterns or structures on its own. When I saw examples of classification and clustering, the concept became much clearer. It helped me see why supervised learning is better for tasks like predicting outcomes, while unsupervised learning is used to discover hidden relationships in data.

Another important part of this lab was model evaluation. I learned that it is not enough to train a model; it's also essential to test how well it performs with new data. I explored metrics like accuracy, precision, and confusion matrices, which showed me how models can be compared in a fair way. Understanding overfitting and underfitting was also a key moment. It helped me see how some models can memorize training data instead of learning real patterns. This part of the lab made me appreciate the importance of validation techniques and crossvalidation in achieving better generalization.

During the lab, I faced some small challenges when interpreting the outputs of some evaluation cells. Sometimes I didn't fully understand what the results meant. To overcome this, I read the comments in the notebook carefully, checked documentation, and compared results between different learning types. Doing that helped me build confidence and develop a more logical understanding of the ML process. It also improved my ability to read Python code and interpret visual outputs like plots and metrics.

In conclusion, this lab gave me a deeper appreciation of the ML workflow and the logic behind different learning types. I now understand that machine learning is a systematic process that goes far beyond coding — it involves understanding data, choosing the right model, and constantly evaluating results. What seemed difficult at the beginning became much clearer through practice. I am now more confident in analyzing ML problems and I am motivated to explore reinforcement learning and model optimization in future labs.

Kendrick Robinson

Introduction

This journal reflects on my journey through the introductory machine learning course, focusing specifically on my evolving understanding of ML types and the often-daunting workflow process. When I first encountered these concepts, they seemed abstract and disconnected from practical reality. This reflection documents how my perspective transformed through theoretical learning and hands-on application, moving from overwhelmed confusion to structured comprehension. I'll explore the pivotal moments that brought clarity and how this learning has reshaped my approach to technical concepts.

Description of Experience

My engagement with machine learning began through our course modules covering the three primary learning types and the six-step ML workflow. The theoretical foundation was immediately followed by a practical project where we built a tomato disease classification system – a supervised learning task that required us to implement the complete workflow from raw data to functional model. This progression from abstract definitions to concrete implementation provided the crucial context I needed to truly understand what machine learning entails in practice.

Personal Reflection

Initial Thoughts and Feelings:

I remember my first reaction to the ML workflow diagram – it felt like staring at a complex recipe where I didn't understand half the ingredients. The six steps seemed like arbitrary hurdles rather than logical progressions. Data preparation sounded tedious, feature selection mysterious, and the strict separation between training and testing data felt unnecessarily complicated. I wondered why we couldn't just use all our data to train the best possible model.

The distinctions between supervised and unsupervised learning also felt academic rather than practical. The definitions were clear enough in theory, but I struggled to understand why the presence or absence of labels made such a fundamental difference in how algorithms worked.

Analysis and Interpretation:

The turning point came when I started connecting these concepts to familiar analogies. Distinguishing supervised from unsupervised learning clicked when I realized it's about whether you have the 'answer key' during training. Supervised learning became like

studying for an exam with a solution manual – you can check your work and learn from mistakes. Unsupervised learning felt like being given a massive collection of puzzle pieces with no picture on the box and being asked to find which pieces naturally fit together.

Connections to Theoretical Knowledge:

The practical project revealed why our theoretical foundations mattered. When our initial tomato disease classifier achieved 95% training accuracy but only 65% testing accuracy, overfitting transformed from an abstract concept to a concrete problem we needed to solve. Applying regularization techniques we'd learned about theoretically actually worked to improve our model's real-world performance. The workflow steps we'd studied as separate concepts suddenly integrated into a coherent system for building reliable models.

Critical Thinking:

Looking back, I recognize several areas where our approach could have been stronger. We focused heavily on technical implementation but could have benefited from deeper domain research about plant pathology. Our feature selection relied heavily on default image features rather than seeking out the specific visual characteristics that human experts use to identify diseases. We also prioritized overall accuracy without sufficiently considering that in agricultural applications, failing to detect a disease (false negative) might be much more costly than incorrectly flagging a healthy plant (false positive).

Discussion of Improvements and Learning

Personal Growth:

This experience has fundamentally changed how I approach complex technical subjects. I've learned to break down overwhelming concepts into manageable components and to seek out practical examples when theoretical explanations fall short. I've developed more patience with the learning process, understanding that true comprehension often comes through iteration rather than immediate mastery. Most importantly, I've cultivated a healthier skepticism – I now instinctively question whether good performance indicates real understanding or just memorization, both in machine learning and beyond.

Skills Developed:

Beyond the technical skills of data preprocessing and model building, I've developed more transferable capabilities:

- Conceptual translation: The ability to connect abstract theories to concrete implementations
- Workflow thinking: Understanding how different stages of a process interconnect and affect final outcomes

- Diagnostic reasoning: Learning to identify why models fail and how to address specific issues rather than just starting over
- Metacognitive awareness: Better understanding of how I learn complex technical concepts most effectively

Future Application:

The systematic approach of the ML workflow has applications far beyond machine learning. I can see using this methodology for research projects in other courses, analyzing data in future employment, or even approaching complex personal decisions. The caution about overfitting translates to any situation where we might be optimizing for the wrong metrics or failing to test our solutions in realistic conditions. The fundamental insight about the difference between memorizing and understanding has implications for how I'll approach studying and problem-solving in all my future academic work.

Conclusion

My journey through machine learning fundamentals has been transformative, moving from confusion to comprehension through a combination of theoretical learning and practical application. The most valuable insights came from connecting abstract concepts to tangible examples and understanding the purpose behind each step of the ML workflow. I've learned that technical proficiency develops not through memorization but through iterative practice and reflection.

Fernando Reyes

Introduction

I will be reflecting on the lab exercise this week. The purpose of the lab was to explore machine learning and make my own model score a higher accuracy score than the prebuilt model. I will explain my process throughout the lab.

Description of Experience or Topic

Machine Learning builds itself on data that is provided to make predictions and identify patterns. There are also 3 main types of Machine Learning. Supervised Learning is trained off labeled data. It has 2 main examples. Classification is predicting categories, and regression is predicting continuous values. In other words, there are inputs and correct answers provided. Unsupervised Learning finds hidden patterns by clustering and using dimensionality reduction. There's only input features for this method. Reinforcement Learning goes through trial and error. For example, an autonomous vehicle's "reward" would not be crashing. The focus was on Supervised Learning along with ML workflow.

Personal Reflection

I initially believed I knew a good amount of this already based on my AI history class, however I was mistaken. Despite learning the material, my experience taught me otherwise. From my experience, I learned how the ML process works, and how it works with different types of data. Primarily, this one was focused on wine quality. In the end, I experimented with different features to score the highest accuracy. It took trial and error, and I tried over 50 times, and I scored the highest accuracy I could find.

Discussion

This lab helped me grow my understanding of Machine Learning principles and processes. I learned how to navigate datasets and understand visualizations. I also learned the three different types of learning, which can all be used in future model testings. It's also important to note that each model has its strengths and weaknesses. It will be up to me to find out which will be preferable depending on the problem.

Conclusion

In conclusion, I learned Machine Learning, the ML workflow, the different types of learning, and what type of data to work with, along with reflecting on my experiences and applying this to real world applications. I will go on to apply this knowledge for future labs when the time comes. I will also keep practicing and making my own datasets for fun to see what I

can do with what I learnt. My experience has been a fun one, and I learned a lot about how machines “think” and formulate patterns.